

Hybrid Multi-Agent Safety Architecture for Autonomous Vehicle Decision Systems

This project extends the original multi-agent design into a **hybrid architecture** that combines:

- **Sequential safety pipeline (temporal flow)**
- **Parallel specialized agents (modular intelligence)**
- **Central decision coordination (accountability + integration)**

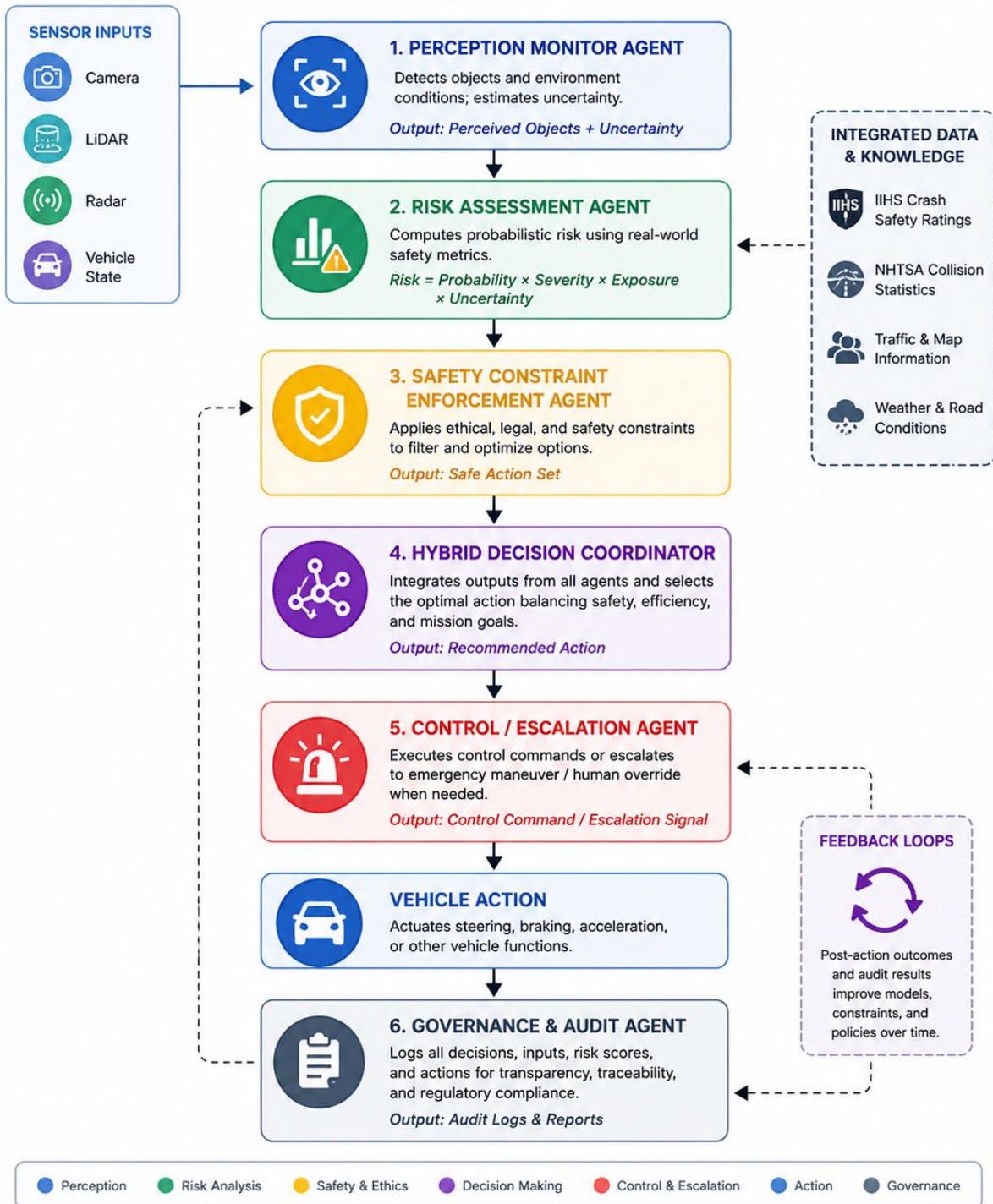
This hybrid approach ensures both **interpretability** and **real-time performance**, addressing limitations of purely centralized or purely decentralized systems.

3.1 Architecture Overview

The system integrates perception, probabilistic risk modeling, ethical constraints, and control decisions through coordinated agents.

Hybrid Multi-Agent Safety Architecture

System Flow



System Flow

Sensor Inputs (Camera, LiDAR, Radar, Vehicle State)



[Perception Monitor Agent]



[Risk Assessment Agent]



[Safety Constraint Enforcement Agent]



[Hybrid Decision Coordinator]



[Control / Escalation Agent]



Vehicle Action



[Governance & Audit Agent]

3.2 Core Agents (Refined Roles)

1. Perception Monitor Agent

- Processes multi-modal sensor inputs
- Performs object detection and scene understanding
- Outputs **uncertainty estimates** alongside predictions
- Detects degraded conditions (low light, occlusion, glare)

2. Risk Assessment Agent

- Computes probabilistic risk scores using:

$\text{Risk} = \text{Probability} \times \text{Severity} \times \text{Exposure}$

- Incorporates uncertainty scaling:

$\text{Adjusted Risk} = \text{Risk} \times \text{Uncertainty Factor}$

- Integrates real-world safety metrics (IIHS, NHTSA)
- Prioritizes high-impact scenarios (e.g., pedestrians, side impacts)

3. Safety Constraint Enforcement Agent

- Applies ethical and safety constraints
- Ensures decisions minimize harm across stakeholders
- Enforces rules such as:
 - Avoid high-severity collision types
 - Prefer braking over risky maneuvering under uncertainty

4. Hybrid Decision Coordinator (NEW – Core Hybrid Component)

- Central component of the hybrid architecture
- Aggregates outputs from all agents
- Resolves conflicts between competing objectives
- Produces a **final decision policy** based on:
 - Risk scores
 - Constraint outputs
 - Uncertainty levels

This component bridges:

- Modular agent intelligence
- System-level accountability

5. Control / Escalation Agent

- Executes final decision
- Triggers emergency actions (braking, steering override)
- Initiates human takeover when necessary

6. Governance & Audit Agent

- Logs all decisions and intermediate outputs
- Maintains traceability across agents
- Supports regulatory compliance and post-hoc analysis

3.3 Hybrid Architecture Rationale

Traditional AV systems fall into two categories:

Fully Centralized Systems

- Single model decides everything
- Fast but lacks interpretability

Fully Decentralized Multi-Agent Systems

- Highly modular
- Difficult to coordinate and audit

Proposed Hybrid System (This Work)

This architecture combines both:

- **Agents handle specialized tasks** (perception, risk, constraints)
- **Central coordinator ensures consistency and accountability**

This results in:

- Improved transparency
 - Better risk-aware decision making
 - Stronger alignment with Responsible AI principles
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3.4 Integration with Temporal Safety Pipeline

The hybrid architecture operates across four phases:

Pre-Collision

- Perception + uncertainty estimation
- Risk computation

Collision Mitigation

- Constraint enforcement
- Decision coordination

Post-Event Recovery

- Control stabilization
- Environment reassessment

Post-Hoc Review

- Governance logging
 - Audit and compliance analysis
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3.5 Example Scenario (Low Visibility Pedestrian)

1. Perception detects pedestrian with high uncertainty
 2. Risk agent inflates risk score due to uncertainty
 3. Constraint agent prioritizes safety (braking over speed)
 4. Decision coordinator selects conservative action
 5. Control agent executes braking
 6. Governance agent logs full decision trace
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3.6 Key Contribution

This hybrid multi-agent framework:

- Embeds Responsible AI principles directly into system structure
- Provides interpretable decision pathways
- Enables quantifiable and auditable safety decisions
- Bridges the gap between modular AI systems and real-world deployment requirements